

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Botany

Paper: BOB-604 — Plant and Microbial Techniques

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following in not more than 50 words each: [10 × 1 = 10]

- (a) What is the role of TCA in separation of protein?
- (b) What is EMBL?
- (c) What is DEAE and ion-exchange chromatography?
- (d) What is DEPC?
- (e) What is the function of Ribonuclease H (RNase H)?
- (f) What are Cy3 and Cy5?
- (g) What is the difference between magnification and resolving power of a lens?
- (h) What is Stokes shift?
- (i) Discuss the surface sterilant in tissue culture.
- (j) What is protoplast fusion?

Question 2. Answer any two of the following: [2 × 7.5 = 15]

- (a) Describe the principle of operation and application of gas chromatography.
- (b) Illustrate the process of PCR with a suitable diagram.
- (c) What is gel filtration chromatography? Explain its merits and demerits.

Question 3. What is a Transmission Electron Microscope (TEM)? Describe the principle and components of TEM with a suitable diagram. [15]

Question 4. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Components and their specific functions of a plant tissue culture medium
- (b) Fluorescence microscopy
- (c) DNA microarray

Question 5. What is micropropagation? Explain the different stages involved in the process with suitable diagrams. Describe its advantages and limitations. [15]

Question 6. What is a cDNA library? How is a cDNA library prepared and what are its applications? [15]

Question 7. Describe any two of the following: [2 × 7.5 = 15]

- (a) Principle and applications of ion-exchange chromatography
- (b) Difference between selective media and indicator media with examples
- (c) Principle of operation and general scheme of a mass spectrometer

Question 8. What is automated DNA sequencing? Describe in detail (a) ILLUMINA and (b) Roche 454, along with their merits and demerits. [15]

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